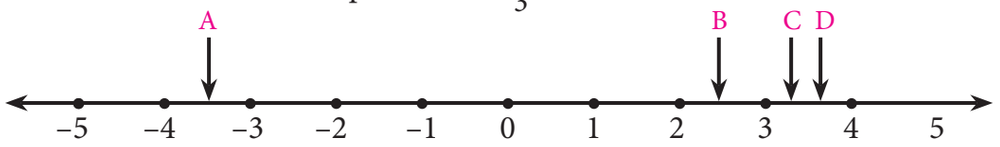
Exercise 2.1

1. Which arrow best shows the position of $\frac{11}{3}$ on the number line?



2. Find any three rational numbers between $\frac{-7}{11}$ and $\frac{2}{11}$.

3. Find any five rational numbers between (i) $\frac{1}{4}$ and $\frac{1}{5}$ (ii) 0.1 and 0.11 (iii) −1 and −2

2.3 Irrational Numbers

You know that each rational number is assigned to a point on the number line and learnt about the denseness property of the rational numbers. Does it mean that the line is entirely filled with the rational numbers and there are no more numbers on the number line? Let us explore.

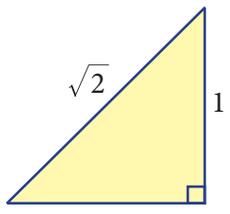


Fig.2.6

Consider an isosceles right-angled triangle whose base and height are each 1 unit long. Using Pythagoras theorem, the hypotenuse can be seen having a length $\sqrt{1^2+1^2} = \sqrt{2}$ (see Fig. 2.6). Greeks found that this $\sqrt{2}$ is neither a whole number nor an ordinary fraction. The belief of relationship between points on the number line and all numbers was shattered! $\sqrt{2}$ was called an irrational number.

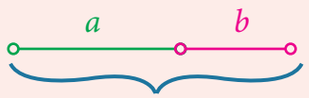
An irrational number is a number that cannot be expressed as an ordinary ratio of two integers.

Examples

1. Apart from $\sqrt{2}$, one can produce a number of examples for such irrational numbers. Here are a few: $\sqrt{5}, \sqrt{7}, 2\sqrt{3}, \ldots$
2. π, the ratio of the circumference of a circle to the diameter of that same circle, is another example for an irrational number.
3. e, also known as Euler's number, is another common irrational number.

GOLDEN RATIO (1:1.6)

The Golden Ratio has been heralded as the most beautiful ratio in art and architecture.



$a+b : a = a : b$

Take a line segment and divide it into two smaller segments such that the ratio of the whole line segment $(a+b)$ to segment a is the same as the ratio of segment a to the segment b.

This gives the proportion $\frac{a+b}{a} = \frac{a}{b}$

Notice that '**a**' is the geometric mean of $a+b$ and b.

4. The golden ratio, also known as golden mean, or golden section, is a number often stumbled upon when taking the ratios of distances in simple geometric figures such as the pentagon, the pentagram, decagon and dodecahedron, etc., it is an irrational number.

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